

DATE : 21.11.2025

DT/NT : DT

LESSON : DATA VISUALIZATION

SUBJECT: INTRO & MATPLOTLIB

BATCH : 322

**DATA
SCIENCE**



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EDUCATION

- Data Visualization Intro
- Matplotlib
- Seaborn
- Plotly
- Visualization project



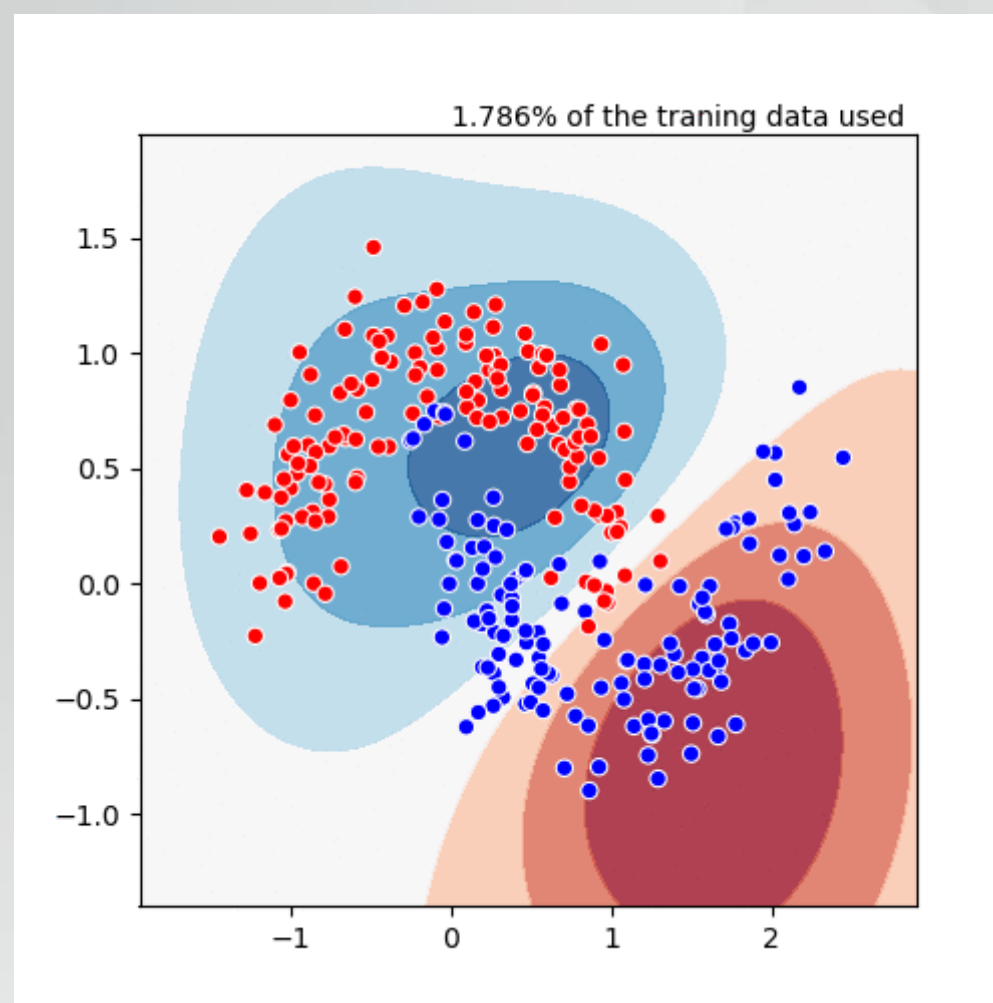
COURSE OVERVIEW

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matplotlib

```
import matplotlib.pyplot as plt
```

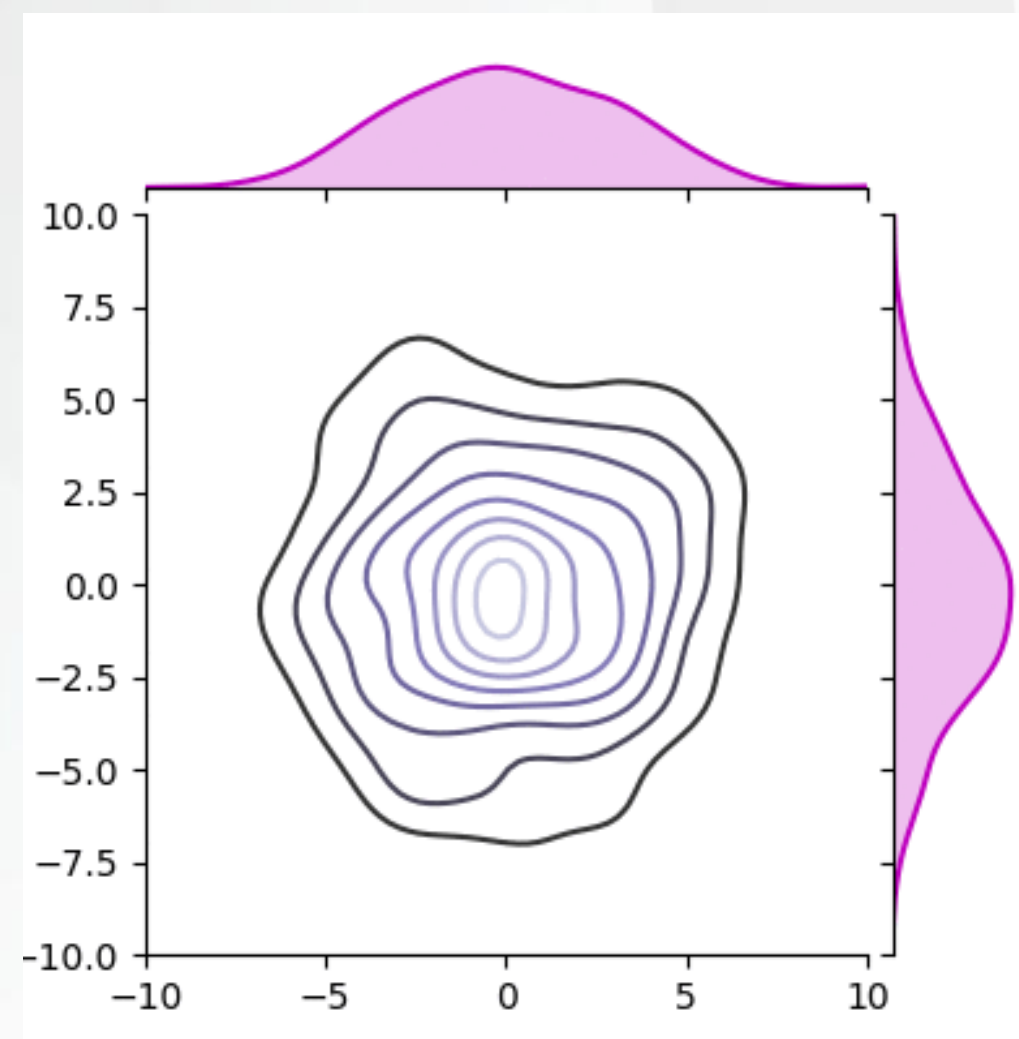
Create figure, axes, subplots

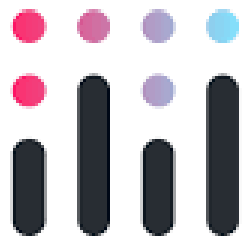


 **seaborn**

```
import seaborn as sns
```

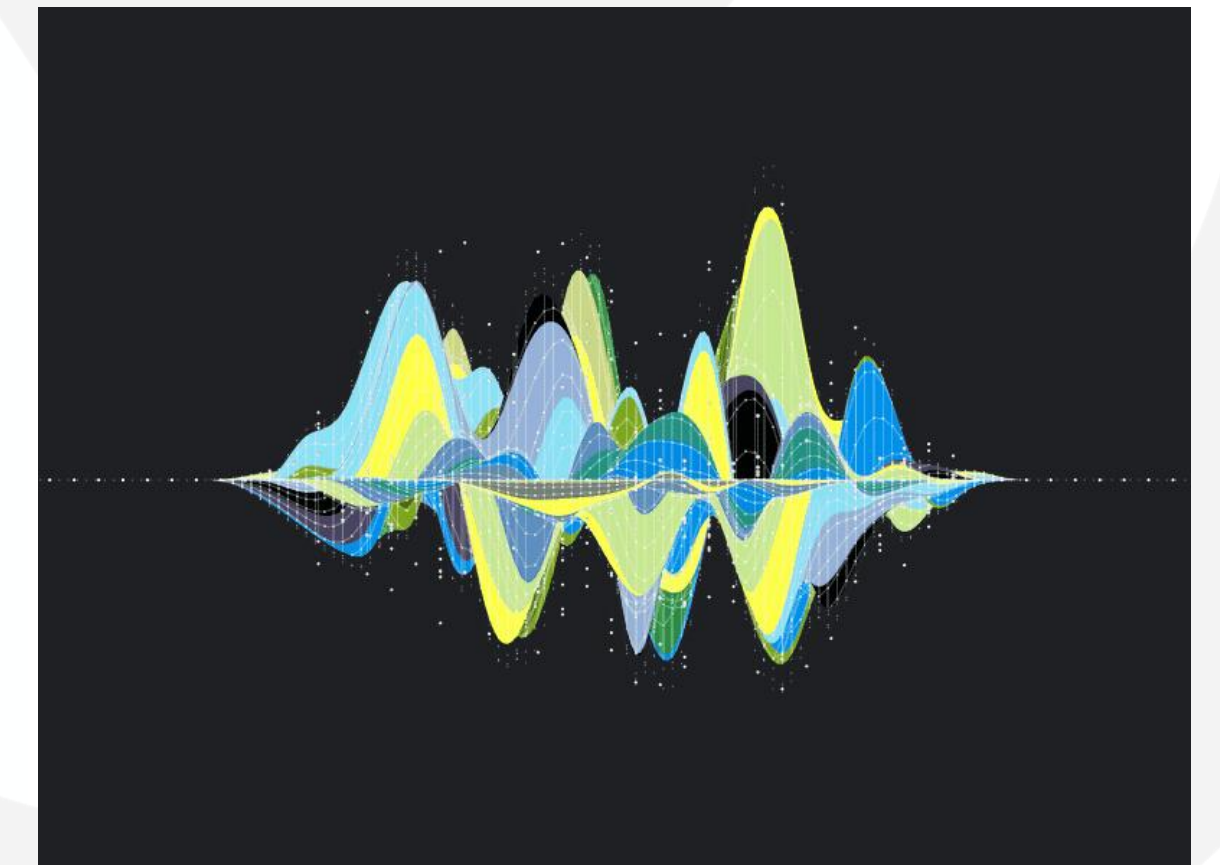
Built on matplotlib and can be used together with it



 **plotly**

```
import plotly as py
```

No need to import matplotlib or seaborn

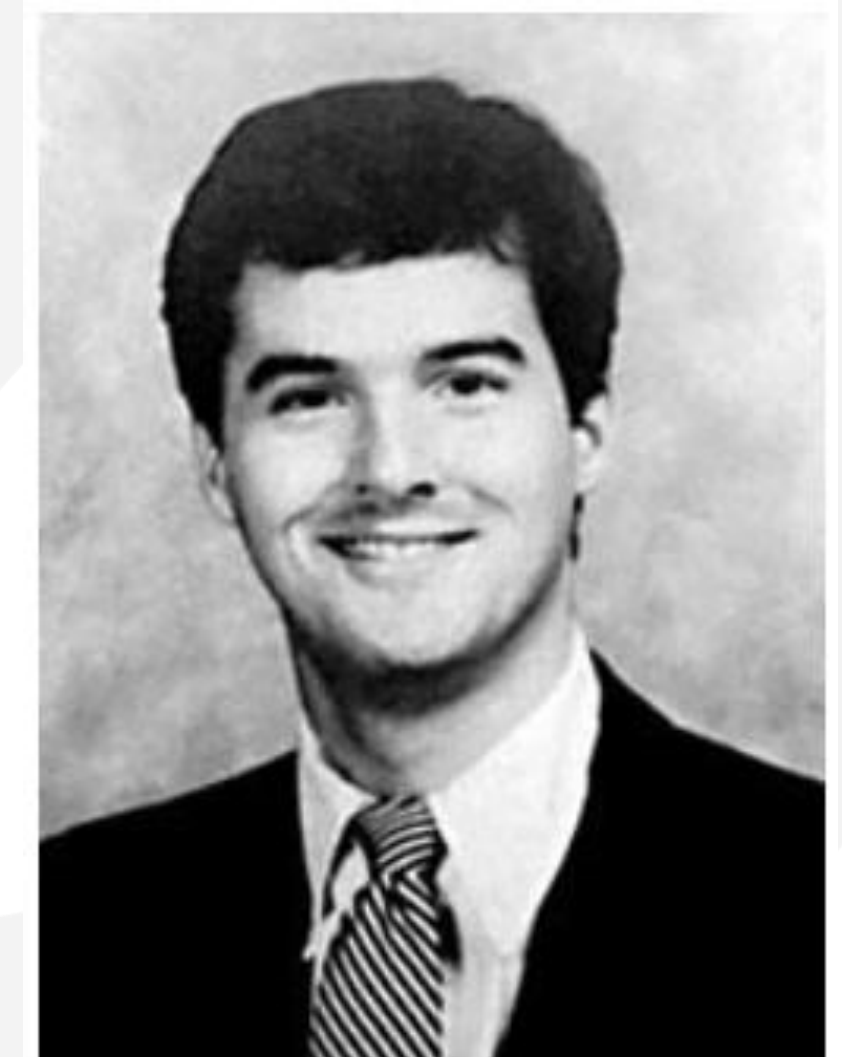




MATPLOTLIB & SEABORN

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- It is used to visualize data in Python.
- In 2002, John Hunter started it as a project to create an interface similar to MATLAB.
- The first version was released in 2003.
- Over time, it was improved, and the Seaborn library was developed from it.





MATPLOTLIB & SEABORN

FEATURES	MATPLOTLIB	SEABORN
Functionality	It is utilized for making basic graphs. Datasets are visualised with graphs styles. • Bar graphs, • Histograms, • Pie charts, • Scatter plots, • Lines and so on.	Seaborn contains a number of patterns and plots for data visualization. It uses fascinating themes. It helps in compiling whole data into a single plot.
Syntax	It uses comparatively complex and lengthy syntax.	It uses comparatively simple syntax which is easier to learn and understand.



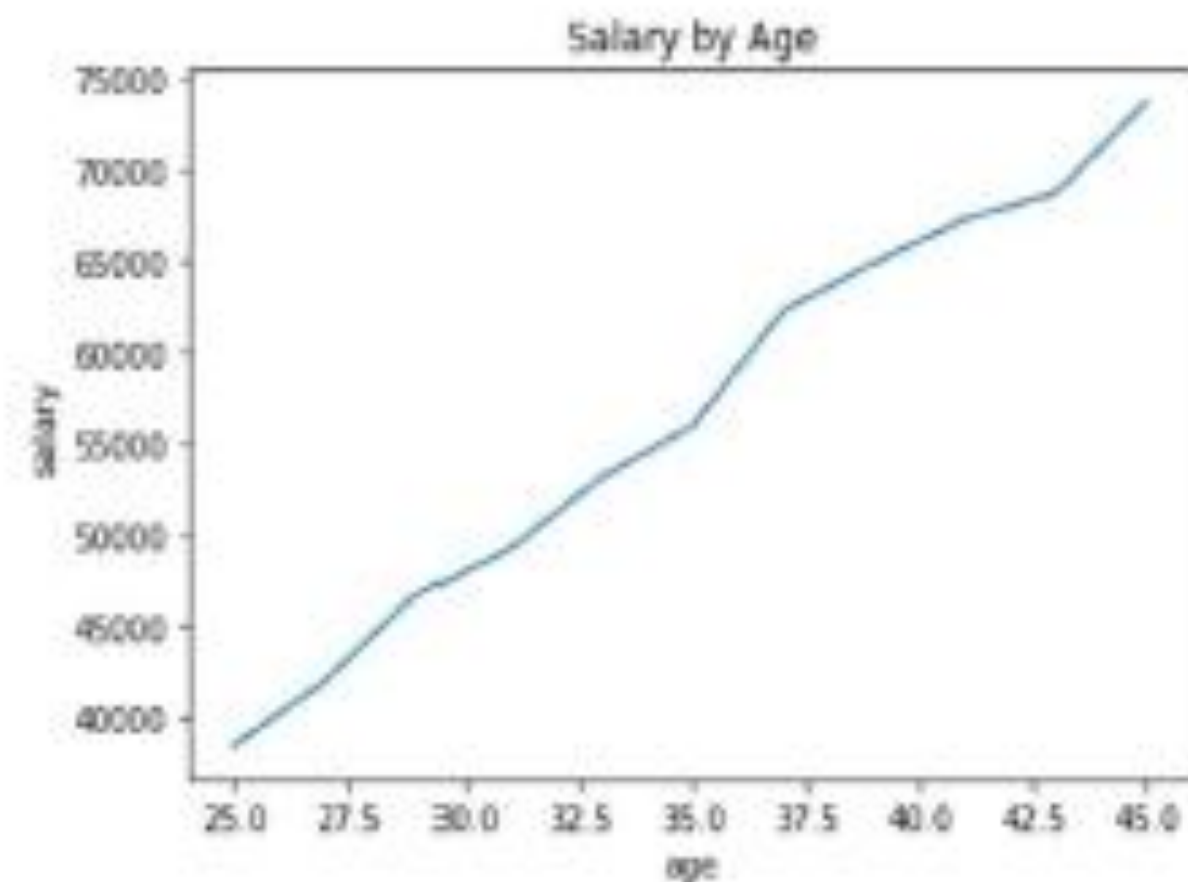
TWO METHODS

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Functional Method

```
plt.plot(age, salary)
plt.xlabel("age")
plt.ylabel("salary")
plt.title("Salary by Age")

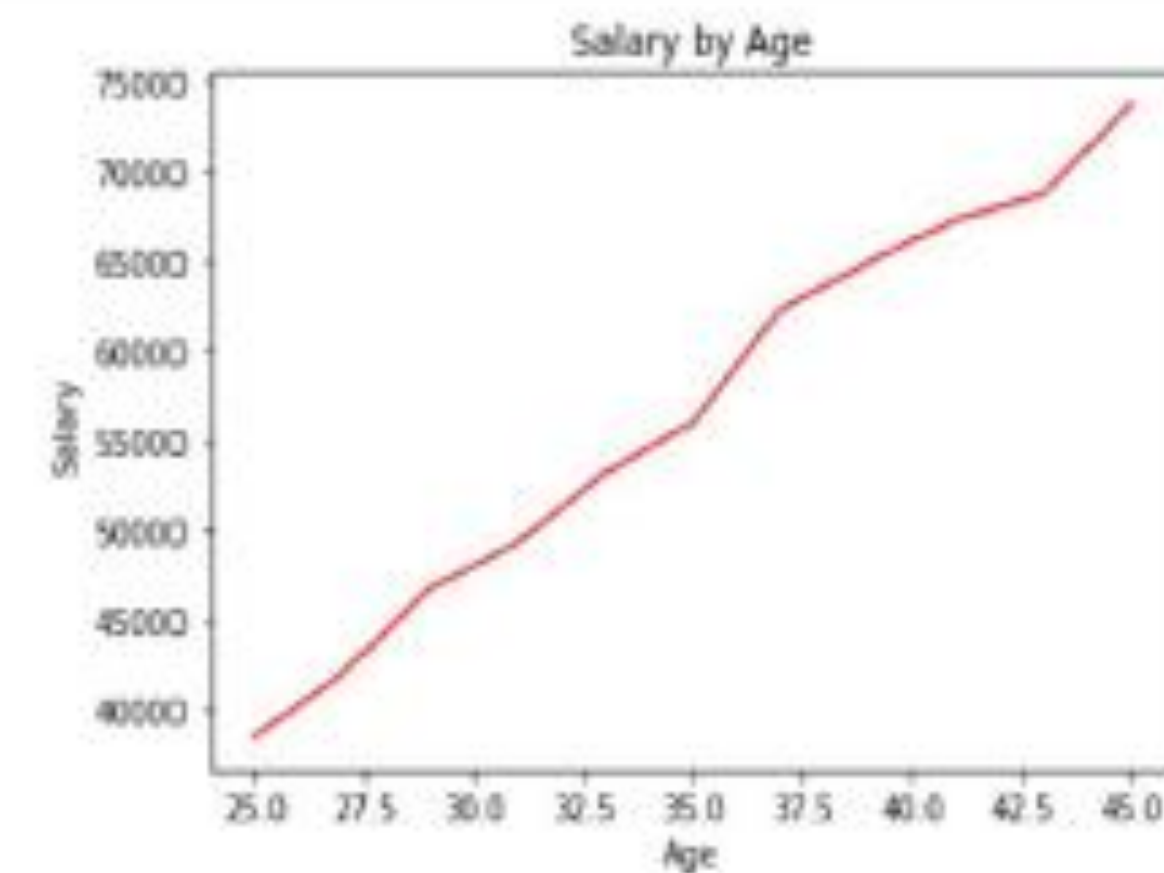
plt.show()
```



Object Oriented

```
fig, ax = plt.subplots()

ax.plot(age, salary, "r")
ax.set_xlabel("Age")
ax.set_ylabel("Salary")
ax.set_title("Salary by Age")
```





AXIS – AXES – FIGURE ?

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Figure, Axes, How should
"Axis" be understood??

matplotlib

```
fig, ax = plt.subplots()
ax = plt.subplot()
ax = plt.axes([l,b,w,h])
```

```
fig = plt.figure()
```

```
ax = fig.add_subplot()
ax = fig.subplots()
ax = fig.add_axes([l,b,w,h])
```

left

width

height

bottom

axes

figure





SCATTER

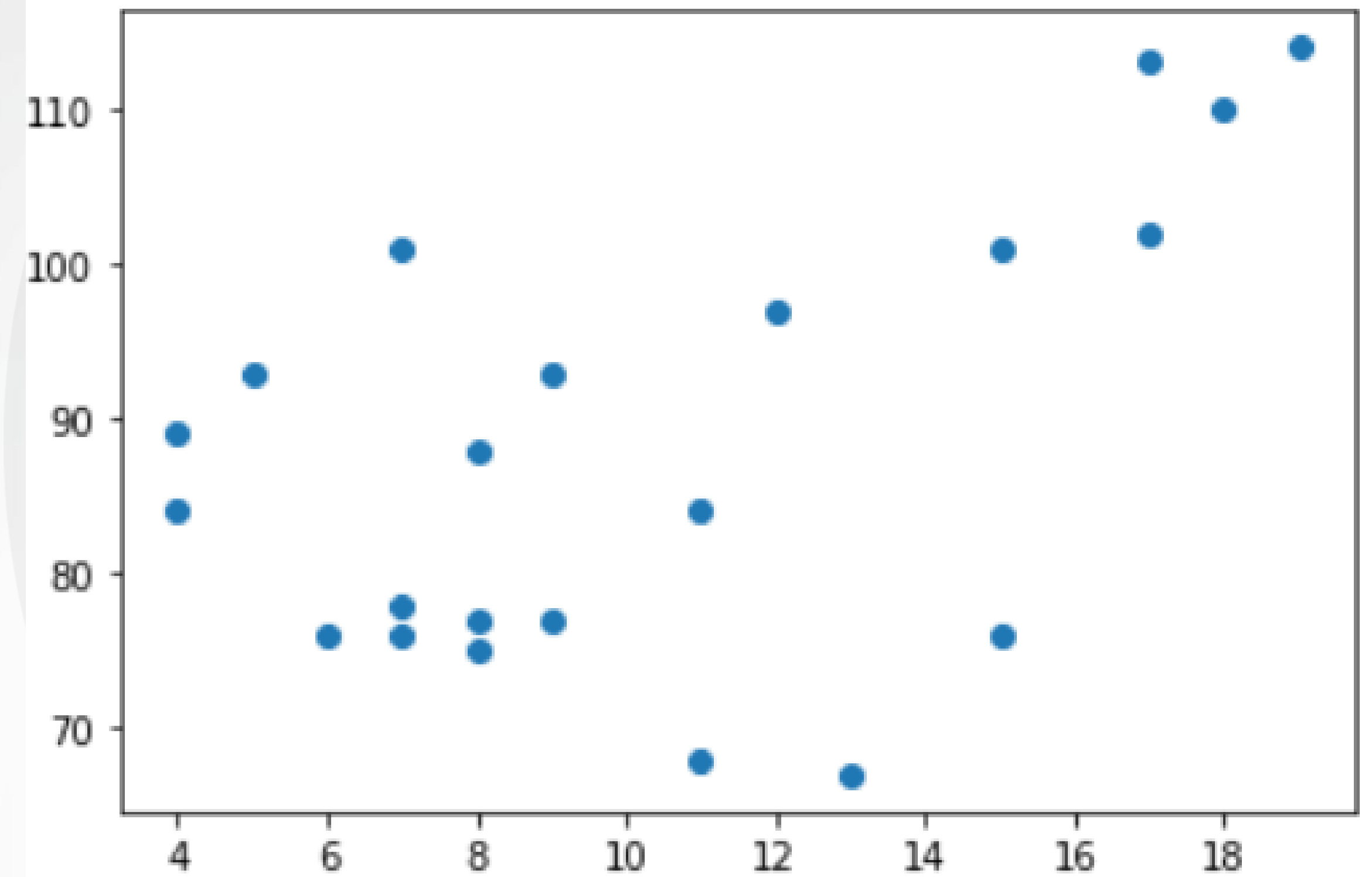
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- Data seti oluşturma

```
1 x = np.array([4, 6, 8, 7, 7, 15, 5 ,9 ,4,  
2             11, 13, 8, 7, 8, 11, 9, 12,  
3             15 ,17 ,17 ,19, 18 ])  
4  
5 y = np.array([89, 76, 77, 78, 101, 76, 93,  
6             77 ,84 ,68, 67, 75, 76, 88,  
7             84, 93, 97, 101 ,102 ,113 ,114, 110 ])
```

- Grafik oluşturma

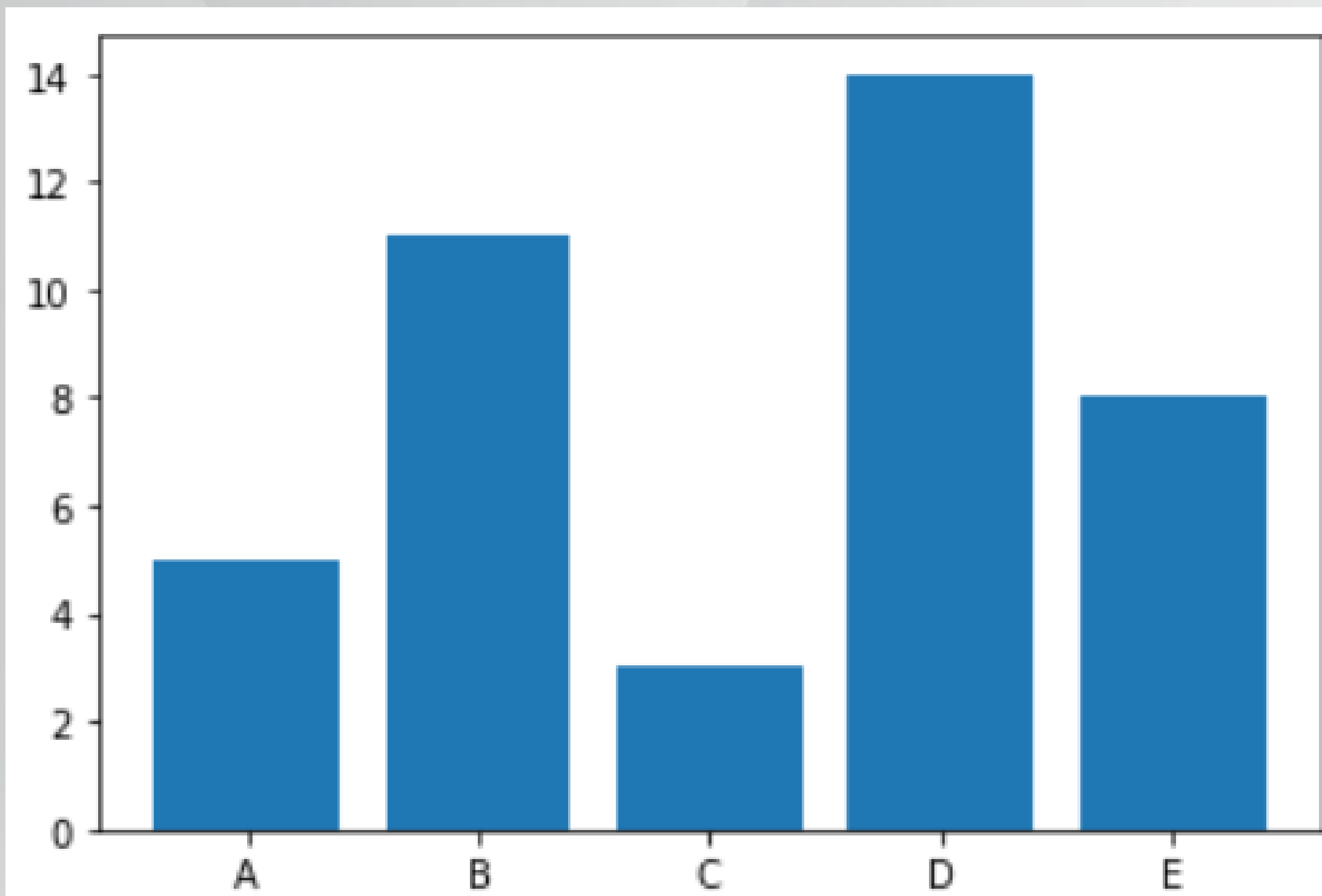
```
1 plt.scatter(x, y)  
2 plt.show()
```





BAR

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- Data seti oluşturma

```
1 x = np.array(["A", "B", "C", "D", "E"])
2 y = np.array([5, 11, 3, 14, 8])
```

- Grafik oluşturma

```
1 plt.bar(x, y)
2 plt.show()
```



HISTOGRAM

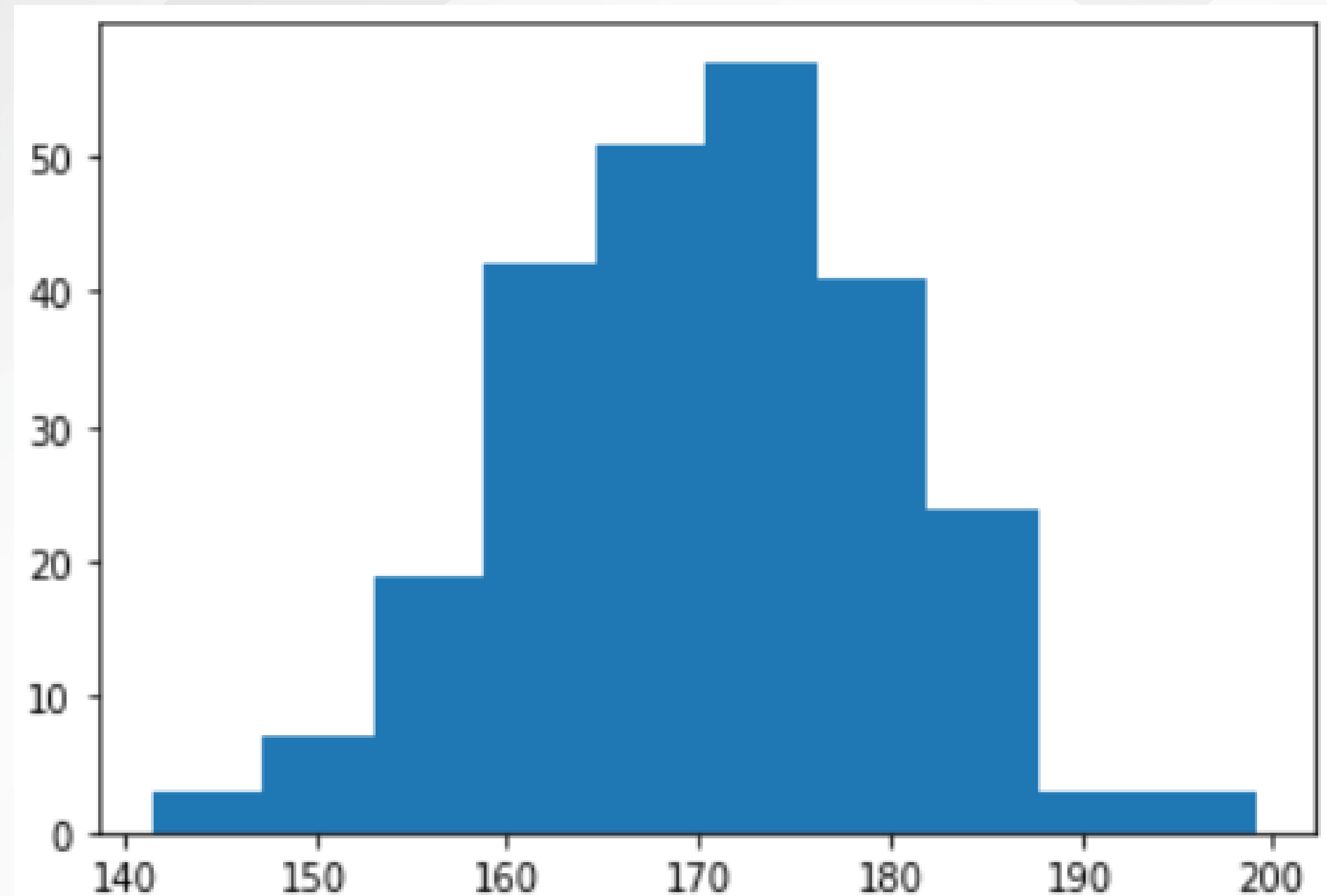
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- Data seti oluşturma

```
1 x = np.random.normal(170, 10, 250)
```

- Grafik oluşturma

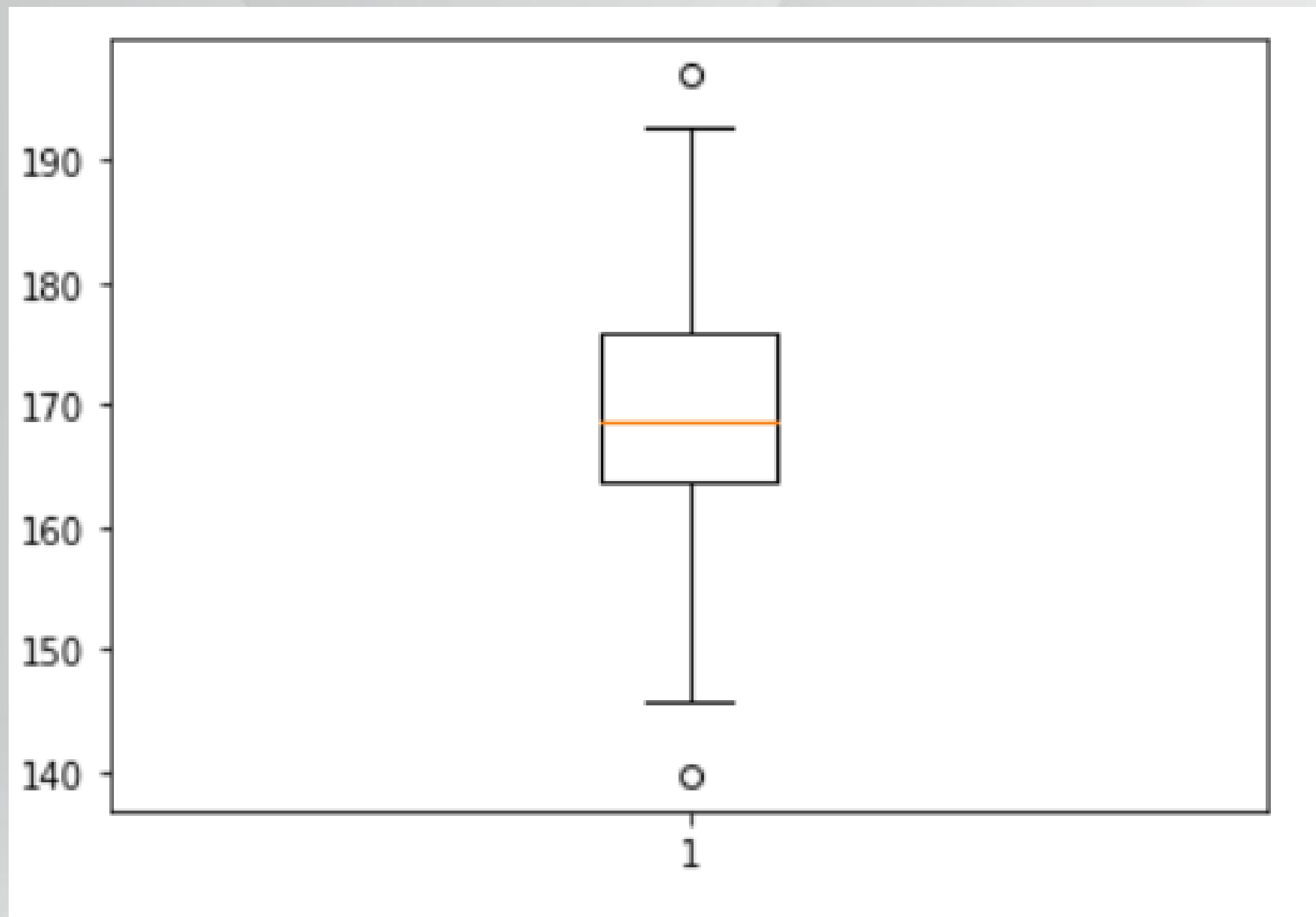
```
1 plt.hist(x)  
2 plt.show()
```





BOX PLOT

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- Data seti oluşturma

```
1 x = np.random.normal(170, 10, 250)
```

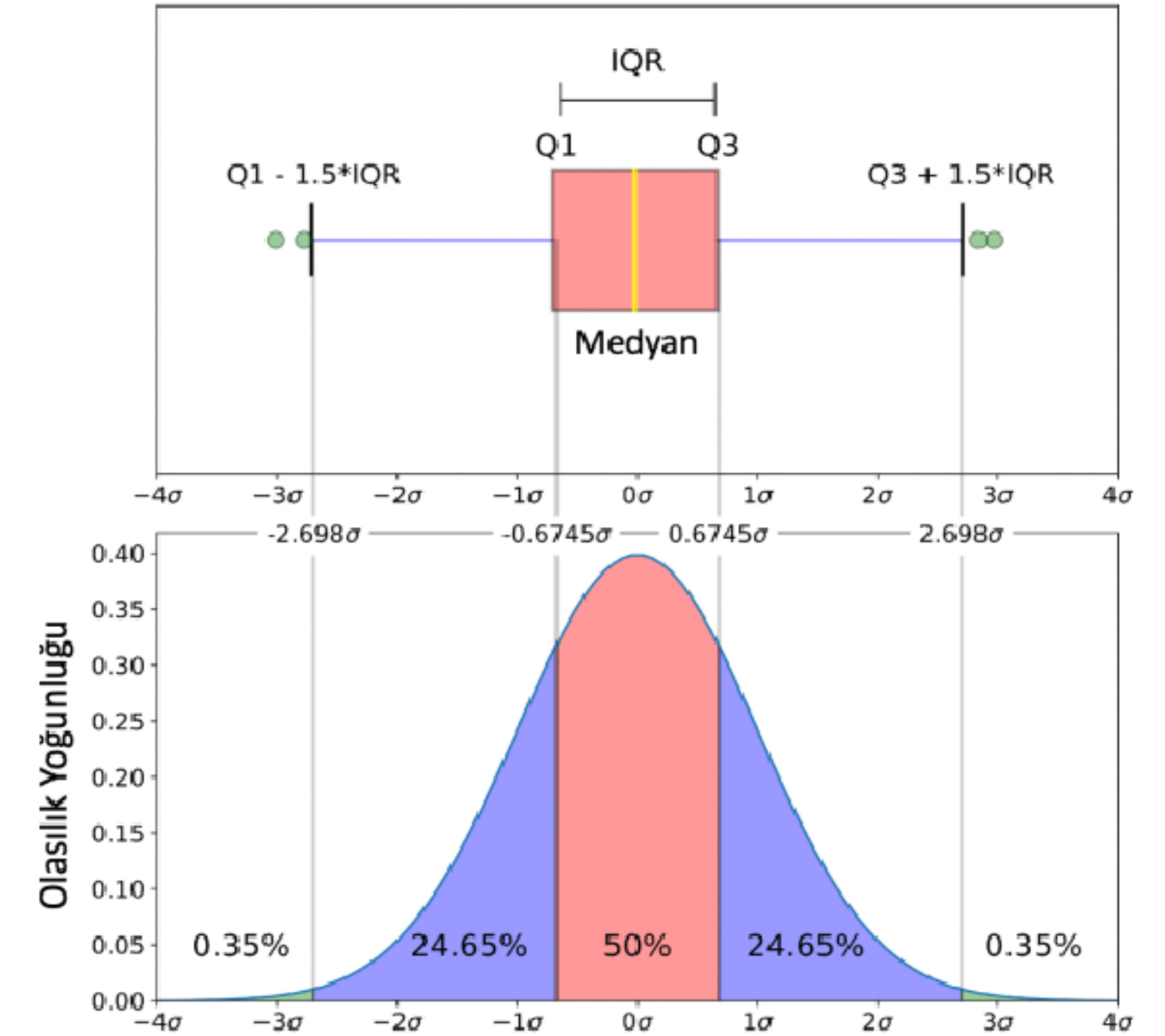
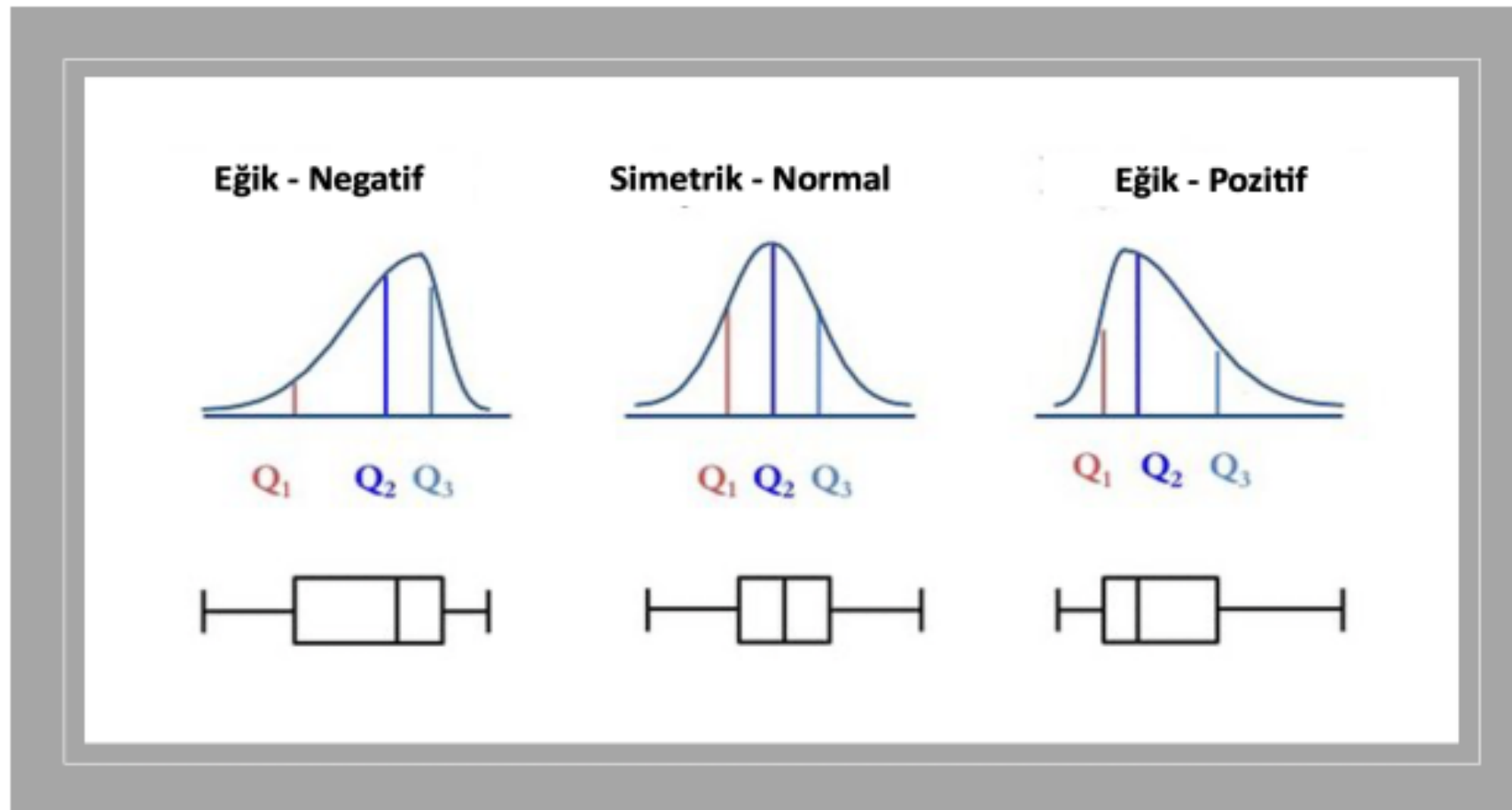
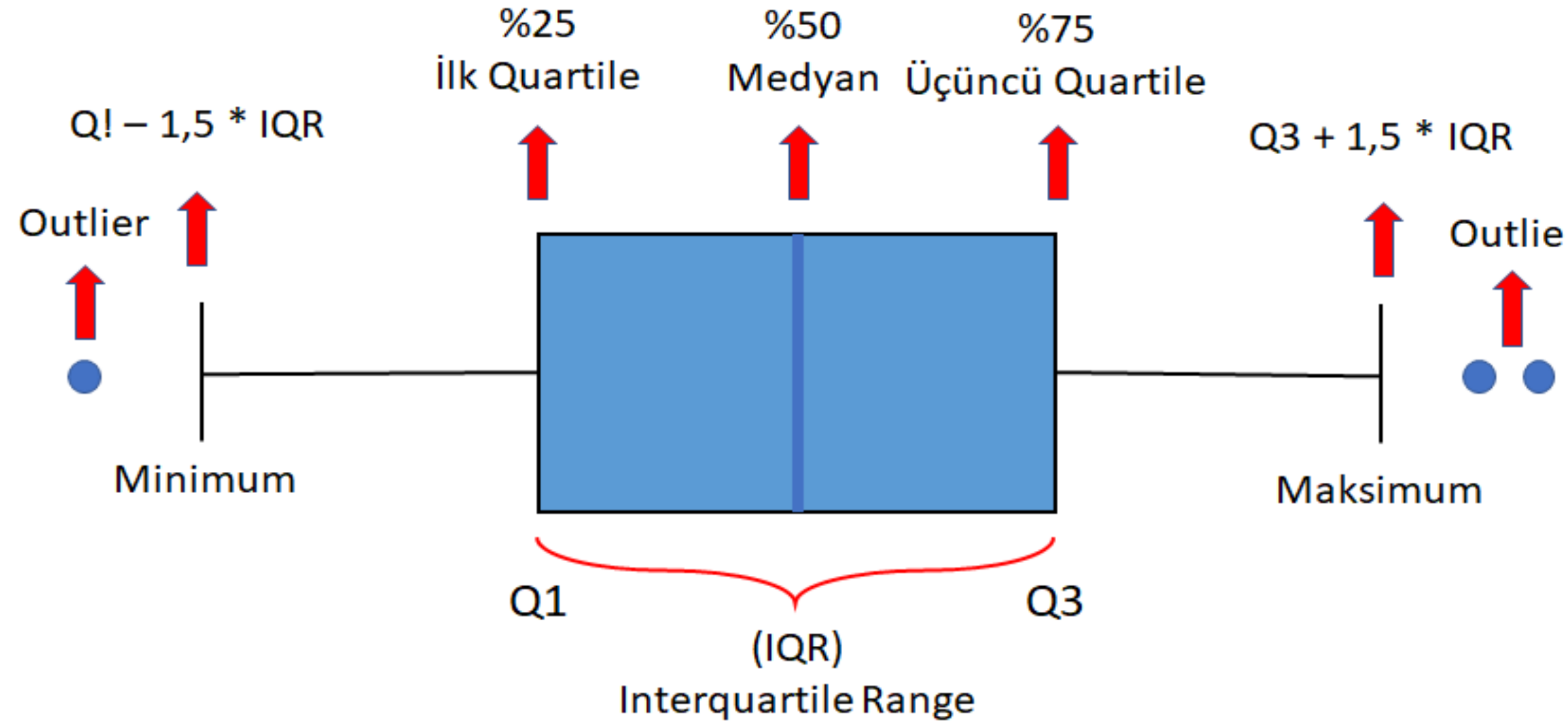
- Grafik oluşturma

```
1 plt.boxplot(x)  
2 plt.show()
```



BOX PLOT

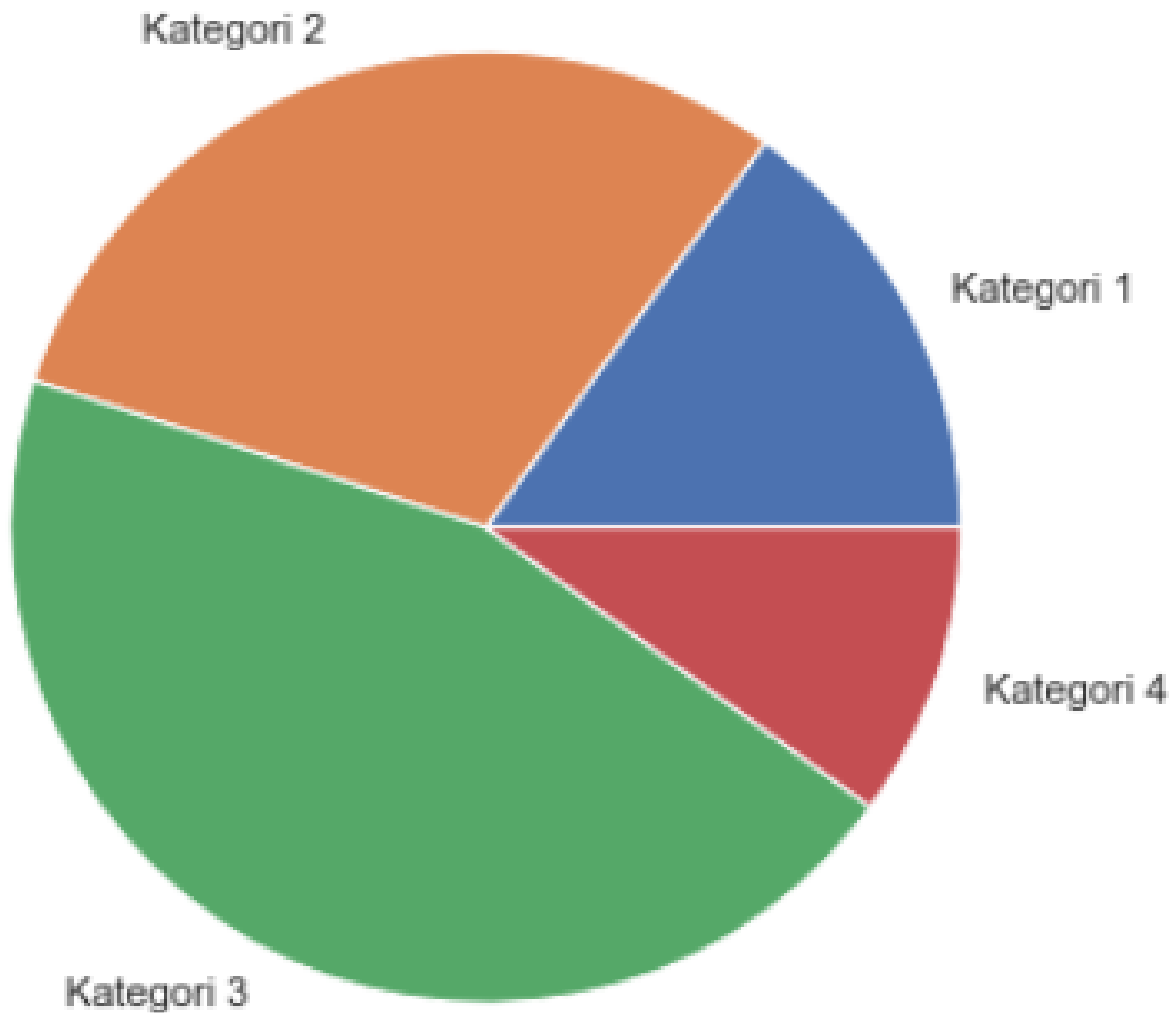
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PIE CHART

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- Data seti oluşturma

```
1 x = ['Kategori 1', 'Kategori 2', 'Kategori 3', 'Kategori 4']
```

- Grafik oluşturma

```
1 import matplotlib.pyplot as plt
2
3 # Veri
4 sizes = [15, 30, 45, 10]
5
6 # Pasta grafiğini oluştur
7 plt.figure(figsize=(6, 6))
8 plt.pie(sizes, labels=x)
9
10 # Göster
11 plt.show()
```